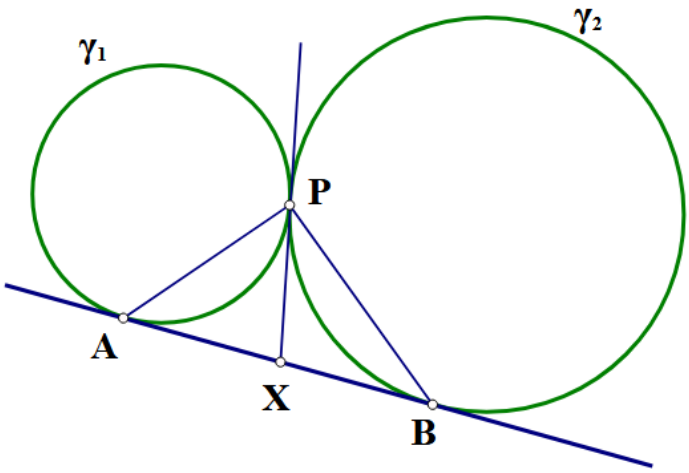
Solutions to Problem Set 1

Solution 1. Let PX be a common tangent to γ_1, γ_2, and denote $\alpha = \angle PAX$, $\beta = \angle PBX$. Since $PX = AX$ and $PX = BX$ because both are tangent from X to γ_1 and γ_2 respectively, triangles PAX and PBX are isosceles, so $\angle APX = \alpha$ and $\angle BPX = \beta$. Then $\angle PAB + \angle APB + \angle PBA = \alpha + (\alpha + \beta) + \beta = 180°$, giving $\angle APB = \alpha + \beta = 90°$.



Solution 2. Let $a > b > 0$ be integers. Set $x = a^2 - b^2 - 2ab$, $y = a^2 + b^2$, $z = a^2 - b^2 + 2ab$. Then $2y^2 = x^2 + z^2$ and $x < y < z$, giving infinitely many such triples. Another solution is to see that $(1, 5, 7)$ is a valid triple and therefore any triple of the form $(t, 5t, 7t)$ with $t > 0$ is also a valid triple.

Solution 3. By Stewart's theorem, $4m_a^2 = 2b^2+2c^2-a^2$ (cyclic). Summing, $4(m_a^2+m_b^2+m_c^2) = 3(a^2 + b^2 + c^2)$, so $\frac{a^2+b^2+c^2}{m_a^2+m_b^2+m_c^2} = \frac{4}{3}$.

Solution 4. The sum of all the factors is zero $\sum_{k=1}^{2n+1}(k-\sigma(k)) = 0$, which is even. If all terms were odd, the sum would be odd. So at least one factor is even, meaning the product is even.

Solution 5. If any three points are collinear, then we are done. So, assume that no three points lie on the same line. Then, take the convex hull of the n points and take three consecutive points on it, say P_1, P_2, and P_n (so that P_1 is in the middle). Now

$$\angle P_2P_1P_n + \angle P_1P_nP_2 + \angle P_1P_2P_n = \pi$$

Because of the construction, notice that

$$\sum_{k=2}^{n-1} \angle P_kP_1P_{k+1} = \angle P_2P_1P_n$$

on account that all the remaining points are inside $\angle P_2P_1P_n$.

Therefore,

$$\pi = \angle P_2P_1P_n + \angle P_1P_nP_2 + \angle P_1P_2P_n = \angle P_1P_nP_2 + \angle P_1P_2P_n + \sum_{k=2}^{n-1} \angle P_kP_1P_{k+1}$$

Thus, at least one of the n angles in the last identity will be less than or equal to π/n, and we are done. □

Solution 6. Since $x = [x] + \{x\}$, then the given inequality can be written as

$$\frac{[x] + 2\{x\}}{[x]^2 + 2\{x\}^2} + \frac{\{x\} + 2[x]}{\{x\}^2 + 2[x]^2} < \frac{4}{[x] + \{x\}}.$$

Putting $a = [x]$ and $b = \{x\}$ we get

$$\frac{a + 2b}{a^2 + 2b^2} + \frac{b + 2a}{b^2 + 2a^2} < \frac{4}{a + b}.$$

Now we prove that

$$\frac{(a + b)(a + 2b)}{a^2 + 2b^2} + \frac{(a + b)(b + 2a)}{b^2 + 2a^2} \leq 4.$$

Indeed,

$$\frac{(a + b)(a + 2b)}{a^2 + 2b^2} = 1 + \frac{3ab}{a^2 + 2b^2} = 1 + \frac{3ab}{(a^2 + b^2) + b^2} \leq 1 + \frac{3a}{2a + b},$$

on account that $a^2 + b^2 \geq 2ab$. Likewise, we get

$$\frac{(a + b)(b + 2a)}{b^2 + 2a^2} \leq 1 + \frac{3b}{a + 2b}.$$

Therefore,

$$\frac{(a + b)(a + 2b)}{a^2 + 2b^2} + \frac{(a + b)(b + 2a)}{b^2 + 2a^2} \leq 2 + 3\left(\frac{a}{2a + b} + \frac{b}{a + 2b}\right).$$

To see that

$$\frac{(a + b)(a + 2b)}{a^2 + 2b^2} + \frac{(a + b)(b + 2a)}{b^2 + 2a^2} \leq 4,$$

it will suffice to prove that

$$3\left(\frac{a}{2a + b} + \frac{b}{a + 2b}\right) \leq 2 \iff 3(a^2+4ab+b^2) \leq 2(2a+b)(a+2b) \iff (a-b)^2 \geq 0.$$

Equality holds when $a = b$. The claimed inequality is strict because there does not exist any positive number x for which $[x] = \{x\}$. □